

CE480 – HW3 Report

Wave Prediction Using ECMWF Wind Data and the JONSWAP Method

Group 3

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Abstract

This report presents the prediction of wave conditions at a selected Black Sea coastal location using hourly ECMWF wind data and the JONSWAP wave growth method. The analysis uses the effective fetch values obtained in HW2. The critical direction is selected as the direction with the maximum effective fetch. A wind rose is plotted to show the distribution of wind speed according to direction. For each year from 2005 to 2025, the maximum wind speed blowing from the critical direction is selected, the storm duration covering this maximum wind is determined, and the JONSWAP method is used to calculate significant wave height and peak wave period. Finally, each year is classified as fully arisen sea (FAS), fetch limited, or duration limited.

Contents

1	Introduction	3
2	Input Data	3
3	Effective Fetch and Critical Direction	3
3.1	Effective Fetch Formula	3
3.2	Effective Fetch Table	4
4	Wind Rose	5
5	Selection of Annual Maximum Wind	5
6	Storm Duration	6
7	JONSWAP Wave Prediction Method	6
7.1	Adjusted Wind Speed	6
7.2	Dimensionless Fetch and Duration	7
7.3	Fully Arisen Sea	7
7.4	Developing Sea Equations	7
7.5	FAS Limit	7
7.6	Classification	8
7.7	Conversion to Physical Wave Parameters	8
8	Results	8
8.1	Annual JONSWAP Results	8
8.2	Discussion	10
9	Conclusion	10

1 Introduction

Wind-generated waves are important in coastal and harbor engineering because they control the design conditions for many coastal structures. Wind waves are mainly controlled by three physical factors:

- wind speed,
- fetch length,
- storm duration.

Fetch is the horizontal distance over which the wind blows on the water surface. For the same wind speed, longer fetch and longer duration generally produce larger wave height and longer wave period. However, wave growth has a physical limit. If both fetch and duration are sufficient, the sea becomes fully developed, also called fully arisen sea (FAS). If the fetch is not enough, the sea is fetch limited. If the storm duration is not enough, the sea is duration limited.

In this homework, the annual wave conditions are estimated using the JONSWAP method.

2 Input Data

The analysis uses hourly wind data for the period:

$$2005 \leq \text{year} \leq 2025$$

The data include:

- wind speed at 10 m elevation, U_{10} , in m/s,
- wind direction, in degrees, measured as the direction from which the wind blows.

The offshore reference point used for the fetch analysis is:

$$(41.934003^\circ\text{N}, 28.069026^\circ\text{E})$$

The effective fetch values used in this report are taken from the HW2 fetch analysis.

3 Effective Fetch and Critical Direction

3.1 Effective Fetch Formula

In HW2, the effective fetch was calculated using three rays for each main direction. The rays were taken at:

$$\theta_{\text{main}} - 7.5^\circ, \quad \theta_{\text{main}}, \quad \theta_{\text{main}} + 7.5^\circ$$

The effective fetch formula is:

$$F_{\text{eff}} = \frac{\sum_i F_i \cos^2 \alpha_i}{\sum_i \cos \alpha_i} \quad (1)$$

where:

- F_{eff} is the effective fetch,
- F_i is the fetch distance measured along ray i ,
- α_i is the angular deviation from the main direction.

Directions with effective fetch smaller than 5 km were excluded.

3.2 Effective Fetch Table

Table 1 gives the effective fetch values used in this homework.

Table 1: Effective fetch values from HW2.

Bearing ($^\circ$)	Effective Fetch (km)	Status
0.0	140.68	VALID
22.5	478.25	VALID
45.0	554.31	VALID
67.5	1074.12	VALID
90.0	597.69	VALID
112.5	219.42	VALID
135.0	103.06	VALID
157.5	54.55	VALID
180.0	27.11	VALID
202.5	0.00	EXCLUDED
225.0	0.00	EXCLUDED
247.5	0.00	EXCLUDED
270.0	0.00	EXCLUDED
292.5	0.00	EXCLUDED
315.0	0.00	EXCLUDED
337.5	83.25	VALID

The maximum effective fetch is:

$$F_{\text{critical}} = 1074.12 \text{ km} \quad (2)$$

at:

$$\theta_{\text{critical}} = 67.5^\circ \quad (3)$$

Therefore, $\theta = 67.5^\circ$ is selected as the critical wave generation direction.

4 Wind Rose

The wind rose was plotted using all hourly wind data from 2005 to 2025. The direction was divided into 16 sectors, each having a width of 22.5° . Wind speed values were grouped into speed classes.

The wind rose is useful because it shows both:

- the frequency of wind from each direction,
- the distribution of wind speed magnitudes in each direction.

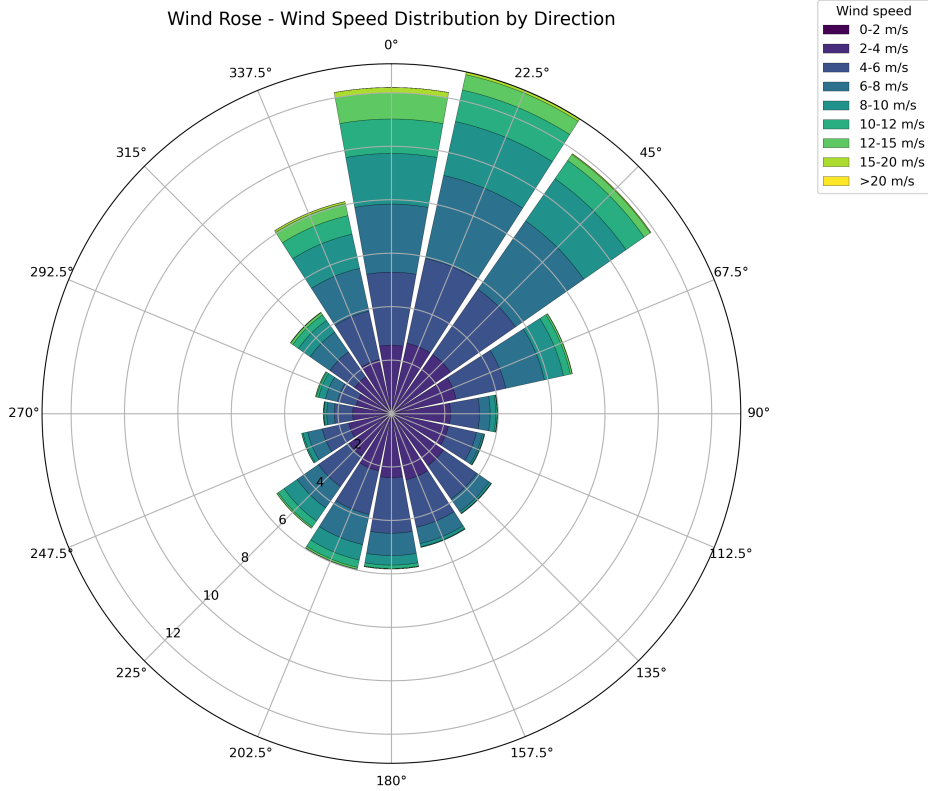


Figure 1: Wind rose for the hourly wind data between 2005 and 2025.

5 Selection of Annual Maximum Wind

For each year, only the winds coming from the critical direction sector were considered. Since the wind rose uses 16 directional sectors, the half-width of each sector is:

$$\frac{22.5^\circ}{2} = 11.25^\circ$$

Therefore, the critical direction sector is:

$$67.5^\circ - 11.25^\circ \leq \theta \leq 67.5^\circ + 11.25^\circ \quad (4)$$

or:

$$56.25^\circ \leq \theta \leq 78.75^\circ \quad (5)$$

For every year, the maximum wind speed $U_{10,\max}$ inside this direction sector was selected.

6 Storm Duration

After the maximum wind speed was found for each year, the storm duration covering that maximum wind was determined.

In this report, the storm is defined as the continuous hourly block around the annual maximum wind where:

1. the wind direction remains inside the critical direction sector,
2. the wind speed remains greater than or equal to a threshold wind speed.

The threshold wind speed is taken as:

$$U_{\text{thr}} = 0.75 U_{10,\max} \quad (6)$$

where:

- U_{thr} is the storm threshold wind speed,
- $U_{10,\max}$ is the yearly maximum wind speed from the critical direction.

The storm duration t is then the number of consecutive hourly records satisfying these conditions. This duration is used in the JONSWAP calculation.

7 JONSWAP Wave Prediction Method

7.1 Adjusted Wind Speed

The wind stress factor, or adjusted wind speed, is calculated as:

$$U_A = 0.71 U_{10}^{1.23} \quad (7)$$

where:

- U_{10} is the 10 m wind speed,
- U_A is the adjusted wind speed.

7.2 Dimensionless Fetch and Duration

The dimensionless fetch is:

$$F^* = \frac{gF}{U_A^2} \quad (8)$$

The dimensionless duration is:

$$t^* = \frac{gt}{U_A} \quad (9)$$

where:

- $g = 9.81 \text{ m/s}^2$,
- F is the effective fetch in meters,
- t is the storm duration in seconds.

The duration-equivalent dimensionless fetch is:

$$F_{\text{dur}}^* = \left(\frac{t^*}{68.8} \right)^{3/2} \quad (10)$$

7.3 Fully Arisen Sea

For fully arisen sea, the dimensionless JONSWAP limits are:

$$H_{m0}^* = 0.243 \quad (11)$$

$$T_p^* = 8.13 \quad (12)$$

7.4 Developing Sea Equations

For developing sea, the dimensionless significant wave height and peak period are calculated as:

$$H_{m0}^* = 0.0016 (F^*)^{1/2} \quad (13)$$

$$T_p^* = 0.286 (F^*)^{1/3} \quad (14)$$

For duration-limited conditions, F_{dur}^* is used instead of F^* .

7.5 FAS Limit

The dimensionless fetch required to reach the FAS height limit is:

$$F_{\text{FAS},H}^* = \left(\frac{0.243}{0.0016} \right)^2 \quad (15)$$

The dimensionless fetch required to reach the FAS period limit is:

$$F_{\text{FAS},T}^* = \left(\frac{8.13}{0.286} \right)^3 \quad (16)$$

Since both height and period should be able to reach the fully arisen condition, the final FAS limit is taken as:

$$F_{\text{FAS}}^* = \max(F_{\text{FAS},H}^*, F_{\text{FAS},T}^*) \quad (17)$$

7.6 Classification

The sea state was classified as follows:

- If $F^* \geq F_{\text{FAS}}^*$ and $F_{\text{dur}}^* \geq F_{\text{FAS}}^*$, the sea is classified as **FAS**.
- If $F^* < F_{\text{dur}}^*$, the sea is classified as **fetch limited**.
- If $F_{\text{dur}}^* < F^*$, the sea is classified as **duration limited**.

7.7 Conversion to Physical Wave Parameters

The physical significant wave height is calculated as:

$$H_{m0} = H_{m0}^* \frac{U_A^2}{g} \quad (18)$$

In this report, H_{m0} is reported as the significant wave height H_s .

The peak wave period is:

$$T_p = T_p^* \frac{U_A}{g} \quad (19)$$

8 Results

8.1 Annual JONSWAP Results

Table 2 gives the yearly maximum wind speed from the critical direction, storm duration, sea state classification, significant wave height, and peak period.

Table 2: Annual maximum wind and JONSWAP wave prediction results for 2005–2025.

Year	Peak Time	$U_{10,\max}$ (m/s)	θ (deg)	Duration (h)	Sea State	H_s (m)	T_p (s)
2005	2005-03-01 22:00	11.95	57.75	10	Duration limited	2.92	8.09
2006	2006-02-05 19:00	11.60	58.36	2	Duration limited	0.83	3.55
2007	2007-02-24 17:00	11.65	57.56	17	Duration limited	4.18	10.39
2008	2008-09-25 22:00	12.51	59.16	11	Duration limited	3.37	8.73
2009	2009-01-06 06:00	12.52	60.66	5	Duration limited	1.87	5.89
2010	2010-03-08 09:00	11.99	63.92	12	Duration limited	3.37	8.88
2011	2011-12-22 12:00	11.83	60.97	14	Duration limited	3.70	9.52
2012	2012-12-19 03:00	14.81	58.36	2	Duration limited	1.22	4.13
2013	2013-11-26 21:00	12.94	57.87	10	Duration limited	3.30	8.50
2014	2014-10-25 05:00	16.30	66.62	11	Duration limited	5.06	10.27
2015	2015-10-08 03:00	13.04	56.76	6	Duration limited	2.28	6.62
2016	2016-10-17 03:00	14.21	67.29	8	Duration limited	3.23	8.05
2017	2017-01-10 13:00	15.97	71.31	6	Duration limited	3.11	7.49
2018	2018-02-26 14:00	14.15	69.38	23	Duration limited	7.07	13.62
2019	2019-11-26 00:00	12.00	73.37	3	Duration limited	1.19	4.44
2020	2020-03-24 06:00	11.95	57.47	11	Duration limited	3.14	8.49
2021	2021-09-06 15:00	12.37	57.86	4	Duration limited	1.55	5.23
2022	2022-02-27 21:00	11.84	60.59	1	Duration limited	0.51	2.54
2023	2023-09-26 22:00	10.04	62.02	2	Duration limited	0.67	3.25
2024	2024-12-01 06:00	12.50	68.26	47	FAS	6.24	13.15
2025	2025-10-02 13:00	14.37	65.83	17	Duration limited	5.78	11.82

8.2 Discussion

The results show that the site is governed by the east-northeast wave generation sector, because the largest effective fetch occurs at 67.5° . The calculated results show that most years are duration limited. This means that the fetch is long enough for large wave growth, but the storm duration is usually not long enough for the waves to become fully developed.

The year 2024 is classified as FAS. This means that both the available fetch and the storm duration were sufficient for the wind to generate waves close to the fully arisen sea condition.

The largest calculated significant wave height in the 2005–2025 period is:

$$H_s = 7.07 \text{ m}$$

which occurred in 2018. The corresponding peak period is:

$$T_p = 13.62 \text{ s}$$

9 Conclusion

This report predicted the wave conditions at the selected Black Sea site using hourly ECMWF wind data from 2005 to 2025 and the JONSWAP method. The effective fetch analysis from HW2 showed that the critical direction is 67.5° , with an effective fetch of 1074.12 km. Therefore, yearly maximum wind speeds from this sector were selected for the wave prediction.

The analysis showed that most annual wave conditions are duration limited. This means that storm duration is usually the controlling factor for wave growth at the site. Only the year 2024 reached the fully arisen sea condition. The largest predicted significant wave height occurred in 2018, with $H_s = 7.07 \text{ m}$ and $T_p = 13.62 \text{ s}$.

Overall, the results show that the site can experience severe waves when strong winds blow from the 67.5° direction for sufficient duration.